

Threads in Java : →

A thread is the smallest unit of processing / execution within a program that can be run independently.

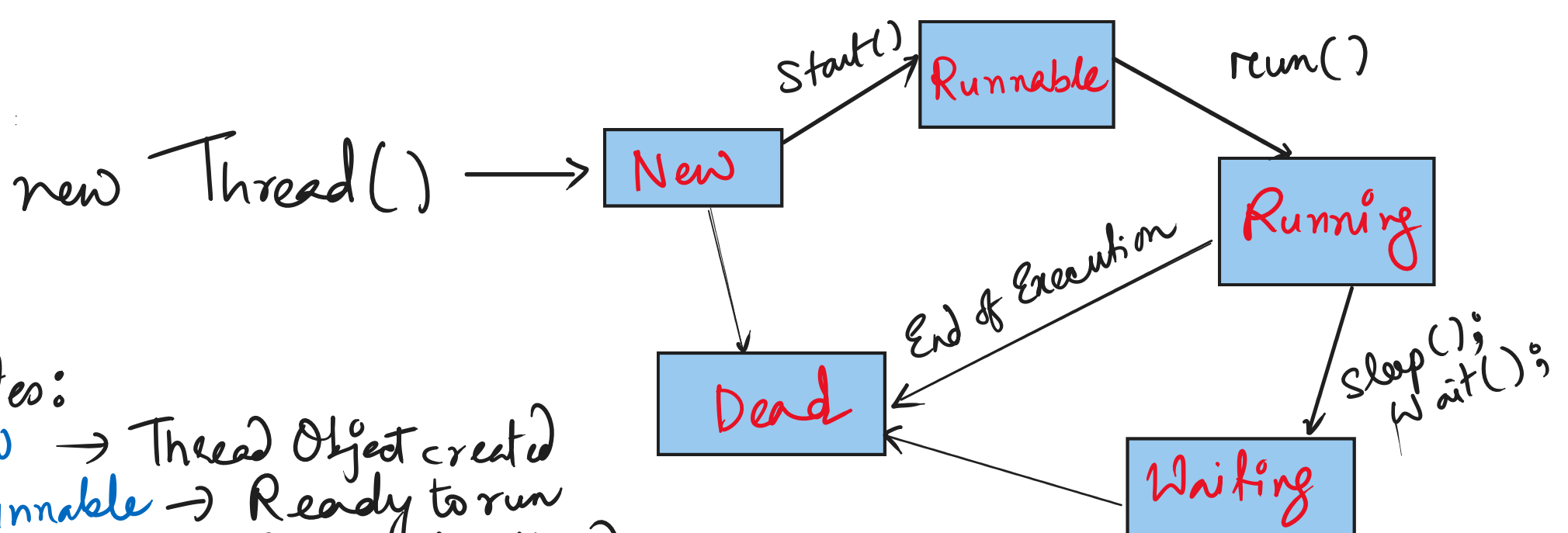
* When a Java program starts, one thread is already running → main thread.

* If your program has only one thread → tasks run one after the other.

* With multiple threads:

* tasks run at the same time (Multi-threading)

Thread Life Cycle : →



States:

- 1 New → Thread Object created
- 2 Runnable → Ready to run
- 3 Running → CPU executing thread
- 4 Waiting / Blocked → Temporarily blocked
- 5 Terminated → Execution Finished

start() vs run()

* start() → creates a new thread

* run() → normal method call (no multi-threading)

* JVM creates a new call stack only when the start() is called.

* Calling run() directly runs on main thread

